

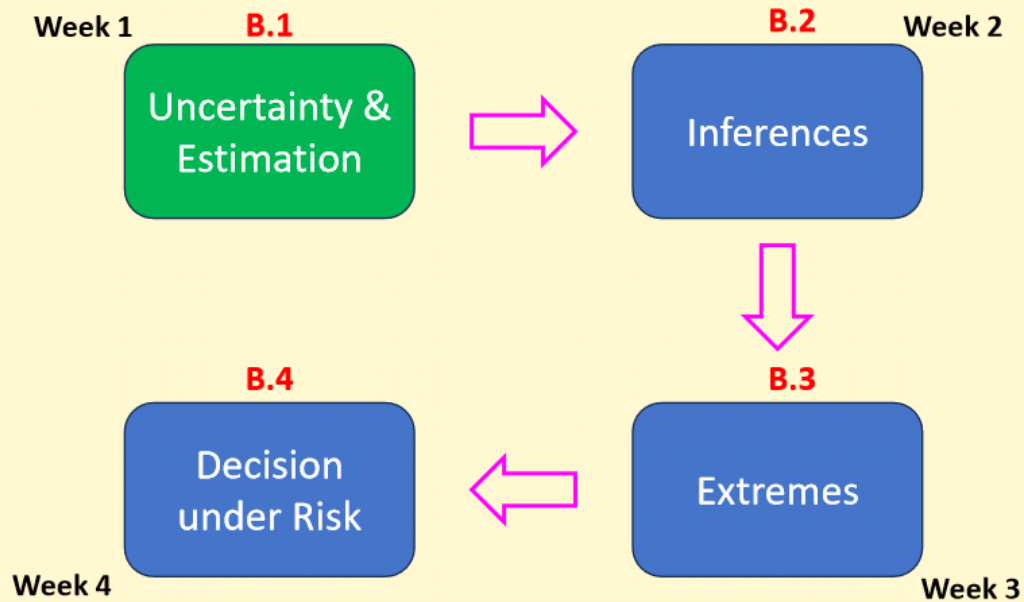


inClass-P2-
L1

CV 5101
Modeling, Uncertainty, and
Data for Engineers
(July – Nov 2026)

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Module Overview



A crash course in probability, probabilistic models, and probabilistic methods.

Check-in with teach-book

<https://mude.citg.tudelft.nl/book/2024/>

- Q1 Topics (Chapters 5, 6, 2, and 3)
 - **Q1C5 Univariate continuous distribution**
 - Q1C5.1 PDF/CDF
 - Q1C5.2 Empirical Distributions
 - Q1C5.3 Parametric Distributions
 - Q1C5.4 Fitting a Distribution
 - **Q1C6 Multivariate Distributions (briefly)**
 - **Q1C2 Propagation of Uncertainty**
 - **Q1C3 Observation Theory: least-sq., Hyp. Test, Conf. Intervals**
- Q2 Topics (Chapter 7 and 8)
 - **Q2C7 Extreme value theory: GEV, return period, POT**
 - **Q2C8 Risk and decision making (CBA)**
- Fundamental Concepts
 - **Chapter 6, 7, 8, 9. Probability basics, rv, z- and t-tables**
- Programming
 - Fundamental Concepts → Chapter 10

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Module Overview

B.1

Uncertainty & Estimation

B.1.1

Probabilistic basics and RVs

- Probability laws
- Random variables
- Discrete, continuous RVs
- PMF/PDF/CDF
- Expectation
- Variance/cov./corr.

B.1.2

Core distributions

- Gaussian
- Uniform
- Exponential
- Lognormal
- Symmetric/skewness
- *Tails*: thick/light

B.1.3

Propagation laws & least-sq.

- Uncertainty propagation
- Least-square errors
- Parameter estimation

B.1.1 Probability basics & random variables

Birthdays!

Birthday-Birthday

Jan 4 14	Feb 22 23	Mar 12 14
Apr 6 15 17	May 6 11 12 14 23	Jun 28 26
Jul	Aug	Sep
Oct	Nov	Dec

Question 1

Probability that
at least two
people in this
room share the
birthday?

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Birthday Problem

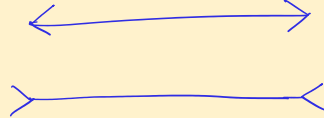
$$\Pr(\text{Two people share birthday}) = 1/365$$

$$\Pr(2 \text{ people share birthday in a class of } 40)$$

$$= \frac{364 \times 363 \times \dots \times (365 - n + 1)}{365 \times 365 \times \dots \text{ 40 times}}$$

for $n=50$, $\Pr(\text{at least 2 shared birthdays}) \approx 0.97$
 $n=23$, $\Pr(\text{at least 2 shared birthdays}) \approx 0.5$

Birthday attack $\leftarrow$ collision attack

$$2^{64} \longrightarrow 2^{32}$$


Intuition can be misleading.

Settlers of Catan



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Practice question

- Work with the [another person around you](#).
- You throw 2 dice;
- add the values of each die
- List all the possible outcomes

Q1. What is the probability that **the sum of the two dice will be 6?**

Q2. What is the probability that you get **at least one 8 in a single round**

consisting of four players? $1 - \left(\frac{31}{36}\right)^4$



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Probability axioms

S : Sample space

$$\Pr(S) = 1$$

mutually
exclusive

E : An event

$$0 \leq \Pr(E) \leq 1$$

E_1, E_2 : Two events s.t. $E_1 \cap E_2 = \emptyset$

$$\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2)$$

Probability rules

$$P_r(E_1 \cup E_2) = P_r(E_1) + P_r(E_2) - P_r(E_1 \cap E_2)$$

Union ↑

$$P_r(\bar{E}_1) = 1 - P_r(E_1)$$

Complement ↑

$$P_r(E_1 \cup E_2 \cup E_3) = P_r(E_1) + P_r(E_2) + P_r(E_3) - P_r(E_1 \cap E_2) - P_r(E_2 \cap E_3) - P_r(E_3 \cap E_1) + P_r(E_1 \cap E_2 \cap E_3)$$

Inclusion -
Exclusion principles

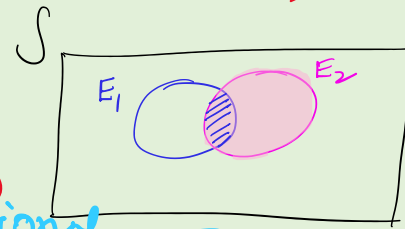
$$\equiv P_r(E_i \cap E_j)$$

$$P_r\left(\bigcup_{i=1}^n E_i\right) = \sum P_r(E_i) - \sum_{i < j} P_r(E_i \cap E_j) + \sum_{i < j < k} P_r(E_i \cap E_j \cap E_k) - \dots + (-1)^{n-1} P_r(E_1 \cap E_2 \cap \dots \cap E_n)$$

Probability rules: conditional prob.

- Conditional probability: $P_r(E_1 | E_2) = \frac{P_r(E_1 \cap E_2)}{P_r(E_2)}$

Probability of E_1 given that E_2 has occurred



- Multiplication rule:

$$P_r(E_1 \cap E_2) = P_r(E_1 | E_2) \cdot P_r(E_2)$$

$$P_r(E_1 \cap E_2 \cap \dots \cap E_n) = P_r(E_1 | E_2 E_3 \dots E_n) \cdot P_r(E_2 | E_3 E_4 \dots E_n) \cdot \dots \cdot P_r(E_{n-1} | E_n) \cdot P_r(E_n)$$

Marginal

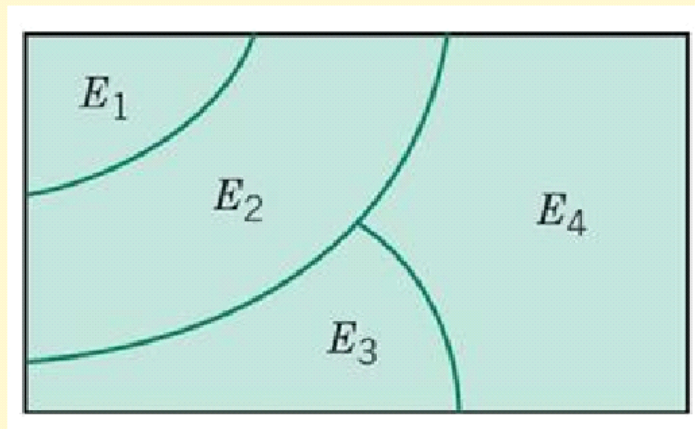
- Two events are called statistically independent (SI), if

$$P_r(E_1 | E_2) = P_r(E_1)$$

$$\Rightarrow \frac{P_r(E_1 \cap E_2)}{P_r(E_2)} = P_r(E_1) \Rightarrow P_r(E_1 \cap E_2) = P_r(E_1) \cdot P_r(E_2)$$

Non-zero unless either $P_r(E_1)$ or $P_r(E_2)$ is zero

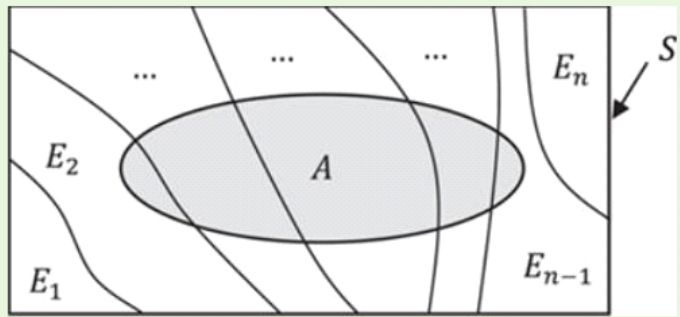
Set Theory: Mutually Exclusive Events



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Probability rules: total probability rule

- If $E_1, E_2, \dots, E_n$ are n MECE events,



(At home) Probability rules: total probability rule

The probability that a building will sustain damage due to an earthquake depends on the intensity of shaking. Suppose the intensity is discretized into the states of weak, moderate, and strong. Assume it is determined that the probability of damage to the building is 0.01, 0.10, and 0.60 for the three levels of shaking intensity, respectively. Furthermore, based on the history of past earthquakes in the region, it is estimated that, for a randomly occurring earthquake, the probabilities for the three levels of shaking intensity at the site of the building are 0.90, 0.08, and 0.02, respectively. We are interested in the probability that the building will sustain damage during the next earthquake in the region.

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Total Probability Rule: Behind the Scenes!

Partitioning Complex Structural Systems

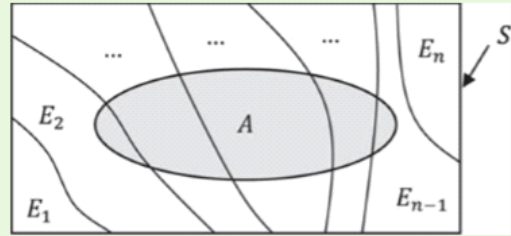
probability that Chennai
Airport is closed tomorrow?

Prob. that a high-rise building
cannot be evacuated safely
during a fire?

Probability rules: Bayes' rule

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- If $E_1, E_2, \dots, E_n$ are n MECE events,



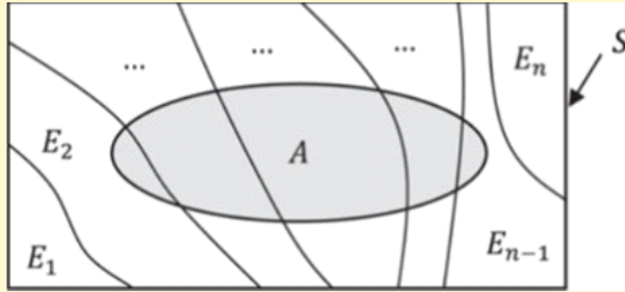
(At home) Probability rules: Bayes' rule

- Assume in the previous example that the building has sustained damage. What can we learn about the intensity of the earthquake?

Total Probability Rules & Bayes' rule

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$E_1, E_2, \dots, E_n$ are (MECE)
partitions of S .



Total Probability and Bayes' Rule